

## Diego A. Zocco

I am an experimental condensed matter physicist with 20+ years of experience working with low temperatures, high magnetic fields, high pressures, and X-ray/neutron scattering techniques. My current research interests focus on quantum materials where the effects of strong electronic correlations and topology are at play.

Nationality: Argentine, Polish (naturalized in 2011)

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Links: [Google Scholar](#), [Web of Science](#), ORCID: [0000-0002-6958-0416](#)



## Education

|      |                                                                                                                                                                                                                                             |
|------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2011 | PhD in Physics, University of California San Diego, USA<br>Advisor: Prof. M. Brian Maple<br>Title: <i>Interplay of Superconductivity, Magnetism, and Density Waves in Rare-Earth Tritellurides and Iron-Based Superconducting Materials</i> |
| 2006 | MS in Physics, University of California San Diego, USA                                                                                                                                                                                      |
| 2003 | MS in Physics, University of Buenos Aires, Argentina                                                                                                                                                                                        |

## Research activities

### Current position and affiliation (since October 2016)

|             |                                                                             |
|-------------|-----------------------------------------------------------------------------|
| Position    | University and Project Assistant; Project Leader (since 01.2019)            |
| Institution | Institute of Solid State Physics, Vienna University of Technology (TU Wien) |
| Address     | Wiedner Hauptstr. 8-10/138, 1040 Vienna, Austria                            |
| Website     | <a href="https://www.tuwien.at/phy/ifp">https://www.tuwien.at/phy/ifp</a>   |

### Previous research appointments

|             |                                                                                                                                                  |
|-------------|--------------------------------------------------------------------------------------------------------------------------------------------------|
| 2011 – 2016 | Postdoctoral Researcher, Karlsruhe Institute of Technology, Germany<br>IQMT New Materials and Thermodynamics group, and Neutron Scattering group |
| 2006 – 2011 | Research Assistant, University of California San Diego, USA                                                                                      |
| 2004 – 2005 | Research Assistant, National High Magnetic Field Laboratory (MPA-NHMFL)<br>Los Alamos National Laboratory, USA                                   |
| 2002 – 2004 | Research Assistant, Physics Department, University of Buenos Aires, Argentina                                                                    |

## Scientific output

- 40+ publications in peer-reviewed international scientific journals and conference proceedings
- h-index = 22 ([Google Scholar](#))
- 20+ invited talks at international conferences/workshops/seminars/colloquia

## Research interests

- Correlation-driven topological insulators and Weyl semimetals.
- Strongly correlated electron systems, quantum phase transitions, heavy fermion compounds, Kondo phenomena, unconventional superconductivity, charge density waves.
- Low temperature physics, high pressures, high magnetic fields, stress/strain dependences, single crystals synthesis and characterization, elastic and inelastic X-ray and neutron scattering

## Funding

- Project title: *Driving spin-orbit coupling to the extreme*  
Agency: Austrian Science Fund (FWF)  
Role: Principal investigator  
Duration: 01.2019 – 06.2023  
Amount: €351,550.50  
Website: [www.ifp.tuwien.ac.at/forschung/projekte/fwf-exsoc](http://www.ifp.tuwien.ac.at/forschung/projekte/fwf-exsoc)

Participated in organizing and writing sections of the following grant proposals and reports:

- Proposals: US DOE-NNSA/SSAA (2008), US DOE (2009), US CDAC Carnegie DOE Alliance Center (2010) (PI: M. B. Maple)
- Reports: US DOE-NNSA/SSAA annual reports (2008, 2009), US DOE final report (2008, 2010), US NSF annual report (2009) (PI: M. B. Maple); Austrian FWF *TopQuantum* annual reports (2016, 2017, 2018) and final report (2019) (PI: S. Paschen)

## Technical expertise

- **Low temperatures:** Extensive experience measuring electrical transport, magnetic, and thermodynamic properties in dilution refrigerators (DR) at various institutions (LANL, UCSD, KIT, TUW) with up to 20 T superconducting magnets, and with commercial  $^3\text{He}$  PPMS and SQUID systems, and custom-built  $^4\text{He}$  refrigerator systems
- **Materials characterization:** Electrical transport, Hall effect, magnetic susceptibility, torque magnetometry, Faraday magnetometry, heat capacity, thermal expansion, magnetostriction, powder and single crystal X-ray diffraction, Rietveld refinement
- **High pressures:** Extensive experience in the preparation of high pressure cells (hydrostatic clamps, Bridgman-anvil, diamond-anvil cells) for measurements of electrical resistivity, ac-magnetic susceptibility, ac-calorimetry (UCSD, TUW), and for elastic and inelastic X-ray scattering experiments at synchrotron user facilities (ESRF, Argonne)
- **High magnetic field laboratories/user facilities:**
  - 1 year full-time appointment at the National High Magnetic Field Lab, Los Alamos, USA
  - Expertise preparing proposals and performing experiments at various dc- and pulsed-field user facilities (Los Alamos, Tallahassee, Nijmegen) for torque magnetometry, magnetotransport, magnetization, dHvA and magnetostriction
- **X-ray & neutron scattering:** 3.5 years full-time appointment at the Karlsruhe Institute of Technology (KIT-IQMT) neutron scattering group
  - Expertise in X-ray and neutron scattering experiments at various user facilities: X-ray synchrotrons at ESRF-Grenoble (elastic and inelastic + high pressures) and APS-Argonne (inelastic + high pressures), neutron sources at LLB-Saclay, NCNR-NIST and SNS-Oak Ridge
  - Installation, optimization and use of a 4-circle X-ray diffractometer (Huber) equipped with Cu and Mo tube sources and graphite and germanium monochromators, for temperature-dependent measurements (5-800 K) with a closed-cycle cryostat (ARS)
- **Materials synthesis:** Single- and poly-crystalline bulk material synthesis: solid state reaction, molten-flux [filled skutterudites  $(\text{Nd,Ce})_{1-x}\text{La}_x\text{Ru}_4\text{Sb}_{12}$ ,  $\text{CuFeTe}_2$ ], Czochralski growth in multi-arc furnace [ $\text{URu}_2\text{Si}_2$ ,  $\text{UCoGe}$ ]; experience with glass bench (quartz/pyrex)
- **Programming and computing experience:** LabView, Python (core, data acquisition); OriginPro, Excel, NumPy, SciPy, Matplotlib, COMSOL, FullProf, Mantis, PyCrystalField, Matlab mview/mfit (data analysis); GitHub, LaTeX, HTML.

## List of recent collaborators

S. Paschen, D. M. Kirschbaum, S. Dzsaber, A. Prokofiev, X. Yan, E. Bauer (TU Vienna); M. B. Maple (UC San Diego); K. Grube, F. Weber, R. Heid, T. Wolf, H. v. Löhneysen (Karlsruhe Institute of Technology); S. Rosenkranz, A. H. Said (Argonne National Laboratory); A. Bossak, G. Garbarino (ESRF, Grenoble); M. Jaime, R. D. McDonald, F. Weickert, V. Zapf (LANL high-field facility); A. McCollam (Nijmegen high-field facility); I. R. Fisher (Stanford); M. G. Vergniory (University of Sherbrooke), E. Svanidze (MPI CPfS); J. Custers (Charles University, Prague); T. Shibauchi (U. Tokyo); Q. Si (Rice University).

## Teaching and supervision activities

### Teaching appointments

|             |                                                                    |
|-------------|--------------------------------------------------------------------|
| 2017 – 2023 | University Assistant, Vienna University of Technology, Austria     |
| 2014        | Teaching Assistant, Karlsruhe Institute of Technology, Germany     |
| 2005 – 2007 | Teaching Assistant, University of California San Diego, USA        |
| 2001 – 2004 | Teaching Assistant/Lecturer, University of Buenos Aires, Argentina |

### Supervision of students

Co-supervised with Prof. Silke Paschen (Vienna): PhD (current), Master's (2021), and Bachelor's (2018) theses of Diana Kirschbaum, Project work of Mathias Pelz (2021), Master's thesis of Lukas Cvitkovich (2019); with Dr. Kai Grube (Karlsruhe): PhD thesis of Felix Eilers (2014); with Prof. M. Brian Maple (San Diego): students Noravee Kanchanavatee, Kevin Huang, Eileen Gonzalez, Nicole Crisosto, and Xiao Chen.

## Other information

### Scholarships and awards

|      |                                                                      |
|------|----------------------------------------------------------------------|
| 2011 | Karlsruhe Institute of Technology Postdoctoral Scholarship, Germany  |
| 2006 | Teaching Assistant Excellence Award, UC San Diego, USA               |
| 2004 | Graduate Scholarship, National Research Council (CONICET), Argentina |
| 2003 | Instituto Balseiro Scholarship, Centro Atómico Bariloche, Argentina  |
| 2002 | MS Dissertation Scholarship, University of Buenos Aires, Argentina   |

### Scientific service

- Regular reviewer for Nature Communications, Physical Review Letters, Physical Review X, Physical Review B, and Proceedings of the National Academy of Sciences
- Reviewer of funding proposals: Deutsche Forschungsgemeinschaft (German Research Foundation, Individual Grants)
- Outreach activities: San Diego Elementary School Science Day 2006, Young Physicists Program 2010 (San Diego, USA), Long Night of Research 2018 (*Lange Nacht der Forschung*, Vienna, Austria)

### Languages

English and Spanish. German: advanced (B2 level).

## Publications

Five selected publications marked with (\*\*\*)

43. (\*\*\*) *Emergent topological semimetal from quantum criticality*  
D. M. Kirschbaum, L. Chen, D. A. Zocco, H. Hu, F. Mazza, M. Karlich, M. Lunik, D. H. Nguyen, J. Larrea Jiménez, A. M. Strydom, D. Adroja, X. Yan, A. Prokofiev, Q. Si, and S. Paschen  
Nature Physics **22**, 218224 (2026), DOI: [10.1038/s41567-025-03135-w](https://doi.org/10.1038/s41567-025-03135-w)
42. (\*\*\*) *The new heavy fermion compound  $Ce_3Bi_4Ni_3$*   
D. M. Kirschbaum, X. Yan, M. Waas, R. Svagera, A. Prokofiev, B. Stöger, P. Rogl, D.-G. Oprea, C. Felser, R. Valentí, G. Vergniory, J. Custers, S. Paschen, and D. A. Zocco  
Phys. Rev. Research **6**, 023242 (2024), DOI: [10.1103/PhysRevResearch.6.023242](https://doi.org/10.1103/PhysRevResearch.6.023242)
41. (\*\*\*) *Control of electronic topology in a strongly correlated electron system*  
S. Dzsaber, D. A. Zocco, A. McCollam, F. Weickert, R. McDonald, M. Taupin, X. Yan, A. Prokofiev, L. M. K. Tang, B. Vlaar, L. E. Winter, M. Jaime, Q. Si, and S. Paschen  
Nat. Commun. **13**, 5729 (2022), DOI: [10.1038/s41467-022-33369-8](https://doi.org/10.1038/s41467-022-33369-8)
40. *A Knudsen cell approach for the molecular beam epitaxy of the heavy fermion compound  $YbRh_2Si_2$*   
E. Bakali, W. Artner, M. Beiser, J. Bernardi, H. Detzce, G. Eguchi, A. Foelske, M. Giparakis, C. Herzig, A. Limbeck, H. Nguyen, L. Prochaska, A. Prokofiev, M. Sauer, S. Schwarz, W. Schrenk, G. Strasser, R. Svagera, M. Taupin, A. S. Thirsfeld, M. Waas, X. Yan, D. A. Zocco, A. M. Andrews, and S. Paschen  
Journal of Crystal Growth **595**, 126804 (2022), DOI: [10.1016/j.jcrysgro.2022.126804](https://doi.org/10.1016/j.jcrysgro.2022.126804)
39. *Anisotropic physical properties of the Kondo semimetal  $CeCu_{1.11}As_2$*   
L. Cvitkovich, D. A. Zocco, G. Eguchi, M. Waas, R. Svagera, B. Stöger, R. Mondal, A. Thamizhavel, and S. Paschen  
JPS Conf. Proc. **30**, 011020 (2020), DOI: [10.7566/JPSCP.30.011020](https://doi.org/10.7566/JPSCP.30.011020)
38. *Electron-phonon coupling and superconductivity-induced distortion of the phonon lineshape in  $V_3Si$*   
A. Sauer, D. A. Zocco, A. H. Said, R. Heid, A. Böhmer, and F. Weber  
Phys. Rev. B **99**, 134511 (2019), DOI: [10.1103/PhysRevB.99.134511](https://doi.org/10.1103/PhysRevB.99.134511)
37. *Competing soft phonon modes at the charge-density-wave transitions in  $DyTe_3$*   
M. Maschek, D. A. Zocco, S. Rosenkranz, R. Heid, A. H. Said, A. Alatas, P. Walmsley, I. R. Fisher, and F. Weber  
Phys. Rev. B **98**, 094304 (2018), DOI: [10.1103/PhysRevB.98.094304](https://doi.org/10.1103/PhysRevB.98.094304)
36. *Determining the local low-energy excitations in the Kondo semimetal  $CeRu_4Sn_6$  using resonant inelastic X-ray scattering*  
A. Amorese, K. Kummer, N. B. Brookes, O. Stockert, D. T. Adroja, A. M. Strydom, A. Sidorenko, H. Winkler, D. A. Zocco, A. Prokofiev, S. Paschen, M. W. Haverkort, L. H. Tjeng, and A. Severing  
Phys. Rev. B **98**, 081116(R) (2018) (Rapid Communications), DOI: [10.1103/PhysRevB.98.081116](https://doi.org/10.1103/PhysRevB.98.081116)
35. *Superconductivity and hybrid soft modes in  $TiSe_2$*   
M. Maschek, S. Rosenkranz, R. Hott, R. Heid, D. A. Zocco, A. H. Said, A. Alatas, G. Karapetrov, S. Zhu, J. van Wezel, and F. Weber  
Phys. Rev. B **94**, 214507 (2016), DOI: [10.1103/PhysRevB.94.214507](https://doi.org/10.1103/PhysRevB.94.214507)
34. *Strain-Driven Approach to Quantum Criticality in  $AFe_2As_2$  with  $A = K, Rb$ , and  $Cs$*   
F. Eilers, K. Grube, D. A. Zocco, T. Wolf, M. Merz, P. Schweiss, R. Heid, R. Eder, R. Yu, J.-X. Zhu, Q. Si, T. Shibauchi, and H. v. Löhneysen  
Phys. Rev. Lett. **116**, 237003 (2016), DOI: [10.1103/PhysRevLett.116.237003](https://doi.org/10.1103/PhysRevLett.116.237003)

33. *Evolution of quasiparticle excitations with critical mass enhancement in superconducting  $A\text{Fe}_2\text{As}_2$  ( $A = \text{K}, \text{Rb}, \text{and Cs}$ )*  
Y. Mizukami, Y. Kawamoto, Y. Shimoyama, S. Kurata, H. Ikeda, T. Wolf, D. A. Zocco, K. Grube, H. v. Löhneysen, Y. Matsuda, and T. Shibauchi  
Phys. Rev. B **94**, 024508 (2016), DOI: [10.1103/PhysRevB.94.024508](https://doi.org/10.1103/PhysRevB.94.024508)
32. *Lattice dynamical properties of superconducting  $\text{SrPt}_3\text{P}$  studied via inelastic X-ray scattering and density functional perturbation theory*  
D. A. Zocco, S. Krannich, R. Heid, K.-P. Bohnen, T. Wolf, T. Forrest, A. Bossak, and F. Weber  
Phys. Rev. B **92**, 220504(R) (2015) (Rapid Communications), DOI: [10.1103/PhysRevB.92.220504](https://doi.org/10.1103/PhysRevB.92.220504)
31. *(\*\*\*) Pressure dependence of the charge-density-wave and superconducting states in  $\text{GdTe}_3$ ,  $\text{TbTe}_3$  and  $\text{DyTe}_3$*   
D. A. Zocco, J. J. Hamlin, K. Grube, J. -H. Chu, H. -H. Kuo, I. R. Fisher, and M. B. Maple  
Phys. Rev. B **91**, 205114 (2015), DOI: [10.1103/PhysRevB.91.205114](https://doi.org/10.1103/PhysRevB.91.205114)
30. *Fermi Surface of  $\text{KFe}_2\text{As}_2$  from Quantum Oscillations in Magnetostriction*  
D. A. Zocco, K. Grube, F. Eilers, T. Wolf, and H. v. Löhneysen  
JPS Conf. Proc. **3**, 015007 (2014), DOI: [10.7566/JPSCP.3.015007](https://doi.org/10.7566/JPSCP.3.015007)
29. *Persistent non-metallic behavior in  $\text{Sr}_2\text{IrO}_4$  and  $\text{Sr}_3\text{Ir}_2\text{O}_7$  at high pressures*  
D. A. Zocco, J. J. Hamlin, B. D. White, B. J. Kim, J. R. Jeffries, S. T. Weir, Y. K. Vohra, J. W. Allen, and M. B. Maple  
J. Phys.: Condens. Matter **26**, 255603 (2014), DOI: [10.1088/0953-8984/26/25/255603](https://doi.org/10.1088/0953-8984/26/25/255603)
28. *Probing the superconductivity of  $\text{PrPt}_4\text{Ge}_{12}$  through Ce substitution*  
K. Huang, L. Shu, I. K. Lum, B. D. White, M. Janoschek, D. Yazici, J. J. Hamlin, D. A. Zocco, P.-C. Ho, R. E. Baumbach, and M. B. Maple  
Phys. Rev. B **89**, 035145 (2014), DOI: [10.1103/PhysRevB.89.035145](https://doi.org/10.1103/PhysRevB.89.035145)
27. *(\*\*\*) Pauli-Limited Multiband Superconductivity in  $\text{KFe}_2\text{As}_2$*   
D. A. Zocco, K. Grube, F. Eilers, T. Wolf, and H. v. Löhneysen  
Phys. Rev. Lett. **111**, 057007 (2013), DOI: [10.1103/PhysRevLett.111.057007](https://doi.org/10.1103/PhysRevLett.111.057007)
26. *Ferromagnetic quantum critical point in  $\text{UCo}_{1-x}\text{Fe}_x\text{Ge}$*   
K. Huang, J. J. Hamlin, R. E. Baumbach, M. Janoschek, N. Kanchanavatee, D. A. Zocco, F. Ronning, and M. B. Maple  
Phys. Rev. B **87**, 054513 (2013), DOI: [10.1103/PhysRevB.87.054513](https://doi.org/10.1103/PhysRevB.87.054513)
25. *High pressure transport studies of the  $\text{LiFeAs}$  analogues  $\text{CuFeTe}_2$  and  $\text{Fe}_2\text{As}$*   
D. A. Zocco, D. Y. Tütüin, J. J. Hamlin, J. R. Jeffries, S. T. Weir, Y. K. Vohra, and M. B. Maple  
Supercond. Sci. Technol. **25**, 084018 (2012), DOI: [10.1088/0953-2048/25/8/084018](https://doi.org/10.1088/0953-2048/25/8/084018)
24. *Intrinsic dependence of  $T_c$  on hydrostatic (He-gas) pressure for superconducting  $\text{LaFePO}$ ,  $\text{PrFePO}$ , and  $\text{NdFePO}$  single crystals*  
N. J. Hillier, N. Foroozani, D. A. Zocco, J. J. Hamlin, R. E. Baumbach, I. K. Lum, M. B. Maple, and J. S. Schilling  
Phys. Rev. B **86**, 214517 (2012), DOI: [10.1103/PhysRevB.86.214517](https://doi.org/10.1103/PhysRevB.86.214517)
23. *High-pressure study of non-Fermi liquid and spin-glass-like behavior in  $\text{CeRhSn}$*   
D. A. Zocco, A. Ślebarski, and M. B. Maple  
J. Phys.: Condens. Matter **24**, 275601 (2012), DOI: [10.1088/0953-8984/24/27/275601](https://doi.org/10.1088/0953-8984/24/27/275601)
22. *Superconductivity, spin and charge order, and quantum criticality in correlated electron materials*  
M. B. Maple, J. J. Hamlin, D. A. Zocco, M. Janoschek, R. E. Baumbach, B. D. White, I. R. Fisher and J.-H. Chu  
EPJ Web of Conferences **23**, 00012 (2012), DOI: [10.1051/epjconf/20122300012](https://doi.org/10.1051/epjconf/20122300012)

21. *High pressure transport properties of the topological insulator  $\text{Bi}_2\text{Se}_3$*   
J. J. Hamlin, J. R. Jeffries, N. P. Butch, P. Syers, D. A. Zocco, S. T. Weir, Y. K. Vohra, J. Paglione, and M. B. Maple  
J. Phys.: Condens. Matter **24**, 035602 (2012), DOI: [10.1088/0953-8984/24/3/035602](https://doi.org/10.1088/0953-8984/24/3/035602)
20. *Twofold enhancement of the hidden-order/large-moment antiferromagnetic phase boundary in the  $\text{URu}_{2-x}\text{Fe}_x\text{Si}_2$  system*  
N. Kanchanavatee, M. Janoschek, R. E. Baumbach, J. J. Hamlin, D. A. Zocco, K. Huang, and M. B. Maple  
Phys. Rev. B **84**, 245122 (2011), DOI: [10.1103/PhysRevB.84.245122](https://doi.org/10.1103/PhysRevB.84.245122)
19. *Low temperature electrical resistivity of praseodymium at pressures up to 120 GPa*  
J. J. Hamlin, J. R. Jeffries, G. Samudrala, Y. K. Vohra, S. T. Weir, D. A. Zocco, and M. B. Maple  
Phys. Rev. B **84**, 033101 (2011), DOI: [10.1103/PhysRevB.84.033101](https://doi.org/10.1103/PhysRevB.84.033101)
18. *Interplay of Superconductivity, Magnetism, and Density Waves in Rare-Earth Tritellurides and Iron-Based Superconducting Materials*  
D. A. Zocco, PhD Dissertation, Department of Physics, University of California San Diego (UMI Dissertations Publishing, 2011), [ProQuest Dissertations Publishing 2011.3466798](https://proquest.com/docview/13466798)
17. *Correlated electron state in  $\text{Ce}_{1-x}\text{Yb}_x\text{CoIn}_5$  stabilized by cooperative valence fluctuations*  
L. Shu, R. E. Baumbach, M. Janoschek, E. Gonzales, K. Huang, T. A. Sayles, J. P. Paglione, J. R. O'Brien, J. J. Hamlin, D. A. Zocco, P.-C. Ho, C. A. McElroy, and M. B. Maple  
Phys. Rev. Lett. **106**, 156403 (2011), DOI: [10.1103/PhysRevLett.106.156403](https://doi.org/10.1103/PhysRevLett.106.156403)
16. *Search for pressure induced superconductivity in  $\text{CeFeAsO}$  and  $\text{CeFePO}$  iron pnictides*  
D. A. Zocco, R. E. Baumbach, J. J. Hamlin, M. Janoschek, I. K. Lum, M. A. McGuire, A. S. Sefat, B. C. Sales, R. Jin, D. Mandrus, J. R. Jeffries, S. T. Weir, Y. K. Vohra, and M. B. Maple  
Phys. Rev. B **83**, 094528 (2011), DOI: [10.1103/PhysRevB.83.094528](https://doi.org/10.1103/PhysRevB.83.094528)
15. *Unconventional T-H Phase Diagram in the Noncentrosymmetric Compound  $\text{Yb}_2\text{Fe}_{12}\text{P}_7$*   
R. E. Baumbach, J. J. Hamlin, L. Shu, D. A. Zocco, J. R. O'Brien, P.-C. Ho, and M. B. Maple  
Phys. Rev. Lett. **105**, 106403 (2010), DOI: [10.1103/PhysRevLett.105.106403](https://doi.org/10.1103/PhysRevLett.105.106403)
14. *Signatures of pressure induced superconductivity in insulating  $\text{Bi}_2\text{Te}_2\text{I}_2$*   
T. Cuk, D. A. Zocco, H. Eisaki, V. Struzhkin, M. Grosche, M. B. Maple, and Z.-X. Shen  
Phys. Rev. B **81**, 184509 (2010), DOI: [10.1103/PhysRevB.81.184509](https://doi.org/10.1103/PhysRevB.81.184509)
13. *The pressure-temperature phase diagram of  $\text{URu}_2\text{Si}_2$  under hydrostatic conditions*  
N. P. Butch, J. R. Jeffries, W. J. Evans, S. X. J. Chi, J. B. Leao, J. W. Lynn, S. V. Sinogeikin, J. J. Hamlin, D. A. Zocco, M. B. Maple  
Mater. Res. Soc. Symp. Proc. **1264**, Warrendale, PA, 2010 (2010 MRS Spring), DOI: [10.1557/PROC-1264-Z10-03](https://doi.org/10.1557/PROC-1264-Z10-03)
12. *Evolution of Magnetic and Superconducting States in  $\text{UCoGe}$  With Fe and Ni Substitution*  
J. J. Hamlin, R. E. Baumbach, K. Huang, M. Janoschek, N. Kanchanavatee, D. A. Zocco, and M. B. Maple  
Mater. Res. Soc. Symp. Proc. **1264**, Warrendale, PA, 2010 (2010 MRS Spring), DOI: [10.1557/PROC-1264-Z12-04](https://doi.org/10.1557/PROC-1264-Z12-04)
11. *Pressure induced superconductivity in the charge density wave compound  $\text{TbTe}_3$*   
J. J. Hamlin, D. A. Zocco, T. A. Sayles, M. B. Maple, J. -H. Chu and I. R. Fisher  
Phys. Rev. Lett. **102**, 177002 (2009), DOI: [10.1103/PhysRevLett.102.177002](https://doi.org/10.1103/PhysRevLett.102.177002)
10. *High-pressure, transport, and thermodynamic properties of  $\text{CeTe}_3$*   
D. A. Zocco, J. J. Hamlin, T. A. Sayles, M. B. Maple, J. -H. Chu and I. R. Fisher  
Phys. Rev. B **79**, 134428 (2009), DOI: [10.1103/PhysRevB.79.134428](https://doi.org/10.1103/PhysRevB.79.134428)



9. *Hydrostaticity and hidden order: effects of experimental conditions on the temperature-pressure phase diagram of  $URu_2Si_2$*   
N. P. Butch, J. R. Jeffries, D. A. Zocco, and M. B. Maple  
High Pressure Res. **29**, Issue 2, 335-343 (2009), DOI: [10.1080/08957950802564676](https://doi.org/10.1080/08957950802564676)
8. *Superconductivity in  $LnFePO$  ( $Ln = La, Pr, \text{ and } Nd$ ) single crystals*  
R. E. Baumbach, J. J. Hamlin, L. Shu, D. A. Zocco, N. Crisosto, and M. B. Maple  
New J. Phys. **11**, 025018 (2009), DOI: [10.1088/1367-2630/11/2/025018](https://doi.org/10.1088/1367-2630/11/2/025018)
7. *New correlated electron physics from new materials*  
M. B. Maple, R. E. Baumbach, J. J. Hamlin, D. A. Zocco, B. J. Taylor, N. P. Butch, J. R. Jeffries, S. T. Weir, B. C. Sales, D. Mandrus, M. A. McGuire, A. S. Sefat, R. Jin, Y. K. Vohra, J. -H. Chu, and I. R. Fisher  
Physica B **404**, 2924 (2009), DOI: [10.1016/j.physb.2009.07.141](https://doi.org/10.1016/j.physb.2009.07.141)
6. *Superconductivity in single crystals of  $LaFePO$*   
J. J. Hamlin, R. E. Baumbach, D. A. Zocco, T. A. Sayles, and M. B. Maple  
J. Phys.: Condens. Matter **20**, 365220 (2008), DOI: [10.1088/0953-8984/20/36/365220](https://doi.org/10.1088/0953-8984/20/36/365220)
5. *Effect of pressure on the superconducting critical temperature of  $La[O_{0.89}F_{0.11}]/FeAs$  and  $Ce[O_{0.88}F_{0.12}]/FeAs$*   
D. A. Zocco, J. J. Hamlin, R. E. Baumbach, M. B. Maple, M. A. McGuire, A. S. Sefat, B. C. Sales, R. Jin, D. Mandrus, J. R. Jeffries, S. T. Weir, and Y. K. Vohra  
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4. *Bose-Einstein Condensation of  $S = 1$  Ni spin degrees of freedom in  $NiCl_2-4SC(NH_2)_2$*   
V. S. Zapf, D. Zocco, B. R. Hansen, M. Jaime, N. Harrison, C. D. Batista, M. Kenzelmann, C. Niedermayer, A. Lacerda, and A. Paduan-Filho  
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Featured in the Journal Club for Condensed Matter Physics, 2005
3. *Approaching field tuned quantum criticality in  $CeIn_{3-x}Sn_x$*   
A. V. Silhanek, D. Zocco, M. Jaime, N. Harrison, T. Ebihara  
Physica B **378–380**, pp. 90-91 (2006), DOI: [10.1016/j.physb.2006.01.036](https://doi.org/10.1016/j.physb.2006.01.036)
2. *Uniaxial pressure dependence of the dynamical properties of vortex lines in  $Bi-2212$  single crystals*  
D. Zocco, M. Monteverde, M. Núñez-Regueiro and C. Acha  
Physica B **354**, pp. 261-265 (2004), DOI: [10.1016/j.physb.2004.09.060](https://doi.org/10.1016/j.physb.2004.09.060)
1. *Uniaxial pressure dependence of the dynamical properties of vortex lines in  $BSCCO$  single crystals*  
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## Selected invited talks at conferences and seminars

- *Correlation and topology in heavy fermion flat band systems*  
APS March Meeting 2025 symposium “Emergent Flat Bands in Bulk Quantum Materials”, Anaheim, USA  
March 18, 2025
- *Correlation-driven topological semimetals*  
International Conference on Superconductivity and Magnetism, Fethiye, Turkey  
April 29, 2024
- *Control of electronic topology in a strongly correlated electron system*  
KITP Conference “Topology, Symmetry and Interactions in Crystals: Emerging Concepts and Unifying Themes”, Santa Barbara, USA  
April 4, 2023
- *Controlling electronic topology in Weyl-Kondo semimetals*  
APS March Meeting 2023 symposium “New Frontiers at the Intersection of Strong Correlations and Topology”, Las Vegas, USA  
March 6, 2023
- *Materials at the intersection of strong electronic correlations and nontrivial topology*  
Institute for Quantum Materials and Technologies, Karlsruhe Institute of Technology, Germany  
January 31, 2023
- *Controlling electronic topology in a strongly correlated electron system*  
Focus Workshop “Topological Materials: From Weak to Strong Correlations”, Dresden, Germany  
April 13, 2022
- *Lattice dynamical properties of  $\text{SrPt}_3\text{P}$  studied via inelastic X-ray scattering and density functional perturbation theory*  
International Conference Superstripes, Ischia, Italy  
June 25, 2016
- *Multiband superconductivity and electronic correlations in  $(\text{K,Rb,Cs})\text{Fe}_2\text{As}_2$  iron-based superconductors*  
International Conference on Superconductivity and Magnetism, Fethiye, Turkey  
April 26, 2016
- *Multiband superconductivity and electronic correlations in iron-based superconductors*  
DFG/GRK 1621 group seminar, Technische Universität Dresden, Germany  
January 12, 2016
- *Electronic correlations in  $\text{AFe}_2\text{As}_2$  iron-based superconductors ( $\text{A} = \text{K, Rb, Cs}$ )*  
International Conference Superstripes, Ischia, Italy  
June 15, 2015
- *Interplay of superconductivity, charge order and magnetism in correlated electron materials*  
Néel Institute seminar, Grenoble, France  
September 23, 2014
- *Paramagnetic Limiting of the Upper Critical Field of Iron-Pnictide Superconductors*  
International Conference on Superconductivity and Magnetism, Antalya, Turkey  
April 29, 2014
- *Exploration of the  $H$ - $T$  phase diagram of  $\text{KFe}_2\text{As}_2$  via thermal expansion and magnetostriction*  
High Magnetic Field Laboratory seminar, Dresden, Germany  
December 6, 2012